$y \sim$

By

$Y \sim N$

$c^T Y \sim N(c^T,$

cmi

Chennai Mathematical Institute

Regression and Classification

Sourish Das

Mid-sem Exam

2nd October 2024

Answer all 6 questions. Write briefly and to the point.
Total Time: 2 hours   Total Marke: 30

1. Show that the least squares method guarantees at least one solution. (3 points)
2. "When there is a high correlation between two predictors, the least squares estimator in a linear regression model becomes unstable and unreliable. - Why? (3 points)
3. (3 points)

   (a) The Ridge estimator for the coefficients of the regression model is defined as

   $$\hat{\beta}_{Ridge} = (X^T X + \lambda I)^{-1} X^T y$$

   find dim

   Show Ridge estimator is a biased estimator?

   (b) If error structure, in linear models, follows $N(0, \sigma^2)$, then find the sampling distribution of the $\hat{\beta}_{Ridge}$.
4. Why LASSO is effective feature selection tool than best-subset selection or forward selection process? (3 points)
5. Write down the following time-series model in linear model format,

   last part

   $$y_t = \beta_0 + \beta_1 y_{t-1} + \epsilon_t, \quad \epsilon_t \sim N(0, \sigma^2), \quad \mathbb{P}(y_0 = 0) = 1, \quad and \quad t = 1, 2, \cdots, T;$$

   and find the OLS estimator for β_0 and β_1. (6 points)
6. Twelve subjects were given oral doses of theophylline then serum concentrations were measured at 11 time points over the next 25 hours. The data is available in `Theoph` dataset available in `datasets` R-package. The datasets cosists of following variables:

   `Dose` dose of theophylline administered orally to the subject (mg/kg).

   `Time`: time since drug administration when the sample was drawn (hr), and

   `conc`: theophylline concentration in the sample (mg/L).

$cov(y_t\, y_{t-1}) = \underline{E(y_t\, y_{t-1})} - E(y_t)\, E(\underline{y_{t-1}) +} -$

$E(y_t^2) E(y) \quad - \overline{E(y_t) E(y_t\, y_{t-1}})$

Following analysis using R is presented below:

```
Call:
lm(formula = log(conc + 1) ~ Time + I(Time^2) + Dose + I(Dose^2),
    data = Theoph)

Residuals:
     Min       1Q   Median       3Q      Max
-1.54738 -0.26046  0.06115  0.41075  0.93670

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.5127840  1.5402729   1.631    0.105
Time         0.1141491  0.0228032   5.006 1.82e-06 ***
I(Time^2)   -0.0058816  0.0009534  -6.169 8.50e-09 ***
Dose        -0.5333977  0.6837919  -0.780    0.437
I(Dose^2)    0.0629102  0.0750044   0.839    0.403
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.583 on 127 degrees of freedom
Multiple R-squared:  0.2647, Adjusted R-squared:  0.2416
F-statistic: 11.43 on 4 and 127 DF,  p-value: 5.892e-08
```

(i) Provide estimate of σ. (3 point)

(ii) If Dose = 4.0, Time = 1.25, then compute expected conc level and 95% Confidence Interval of the conc level. (3 points)

(iii) Which predictor has strongest influence on conc level and why? (3 points)

(iv) What Adjusted R-squared explain with respect to model? (3 points)

$X_0 = (1, 1.25, (1.25)^2, (4.0), (4.0)^2)_{1\times 5}$ $[\quad]_{5\times 5}$ 5×1

$\log(\text{conc}+1) = X_0\beta$

$\text{cons} = e^{X_0\beta} - 1$

$V(\log(\text{conc}+1))$

$y = ax+b$

$\bar{y} = a\bar{x}+b$

$b = \bar{y} - a\bar{x}$

$\cancel{E(y)}$ - ..

conc - .

$\hat{y} - \frac{V(\hat{y})}{V(\log(\text{conc}+1))}$

$V(\text{conc})$

$X_{new}\,\sigma X^T$

Chennai Mathematical Institute

Predictive Analytics: Regression and Classification

Final Exam | 2024-11-23

Total Marks: 50
Total Time: 2 hours

Instructions:
1. Attempt all questions.
2. Write clearly and provide detailed reasoning for all answers.
3. For application questions, interpret the provided R output carefully and justify your conclusions.

Question 1: (10 points)

a) (3 points) Explain the advantages and limitations of the logit link function in logistic regression. Provide examples where this function is particularly useful.

b) (3 points) Derive the relationship between the log-likelihood function and the cross-entropy loss for a binary classification problem.

c) (4 points) Define the concept of basis expansion in regression. Explain its role in addressing non-linearity in data.

Question 2: (10 points)

a) (5 points) Explain the assumptions of linear discriminant analysis (LDA) and discuss why these assumptions may or may not hold in real-world datasets.

b) (5 points) Describe the Gaussian process regression model and explain its limitations when applied to high-dimensional data.

Question 3: (10 points)

a) (3 points) Explain how regularisation techniques (e.g., L1 and L2) influence the performance of regression models.

b) (3 points) Briefly discuss the differences between logistic regression and neural networks in terms of their mathematical formulation and practical application.

c) (4 points) Prove or disprove: The solution to ridge regression minimises the sum of squared errors plus a penalty term on the norm of the coefficients.

$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 \ldots$

log odds

$= 2.5 \times 10^2$

≈ 250

Question 4: (10 points)

An auto manufacturer is analysing factors that influence whether customers purchase an extended warranty. The logistic regression model output in R is provided below:

```
Call:
glm(formula = Purchase ~ Age + Income + Gender, family = binomial, data = auto_data)

Coefficients:
             Estimate  Std. Error  z value  Pr(>|z|)
(Intercept)  -2.456    0.345       -7.115   < 0.001 ***
Age           0.058    0.012        4.833   < 0.001 ***
Income        0.005    0.001        5.000   < 0.001 ***
GenderMale   -0.742    0.215       -3.451   < 0.001 ***
---
Signif. codes:  0 '***' 0.001
```

2.5×10^2

2.5×100

250

$= \frac{1}{1+e^{-(2.5\times10^2)}}$

Answer the following:

a) (3 points) Interpret the coefficients for Age, Income, and GenderMale.

calc

b) (3 points) If a 35-year-old male with an income of $50,000 is evaluated, what is the log-odds of purchasing an extended warranty?

c) (4 points) Calculate the probability of purchase for this individual and explain its implications for the company's marketing strategy.

Question 5: (10 points)

A defense manufacturer is modeling the relationship between production costs (in millions) and factors such as the number of units produced, the complexity of design (measured on a scale of 1 to 10), and the experience of the workforce (in years). The regression output in R is provided below:

```
Call:
lm(formula = Cost ~ Units + Complexity + Experience, data = defense_data)

Coefficients:
             Estimate  Std. Error  t value  Pr(>|t|)
(Intercept)  15.236    2.045       7.448    < 0.001 ***
Units         0.053    0.012       4.417    < 0.001 ***
Complexity    1.672    0.320       5.225    < 0.001 ***
Experience   -0.428    0.134      -3.194    0.002 **
---
Residual standard error: 1.45 on 47 degrees of freedom
Multiple R-squared: 0.785, Adjusted R-squared: 0.765
F-statistic: 39.11 on 3 and 47 DF,  p-value: < 0.001
```

5.1×10^1

Answer the following:

a) (3 points) Interpret the coefficients for Units, Complexity, and Experience.

b) (3 points) Predict the production cost for a project involving 500 units, complexity of 8, and a workforce with 10 years of experience. c) (2 points) Provide 95% predictive interval for your prediction for the same project. d) (2 points) Discuss the implications of the adjusted R-squared value and the significance of the predictors for decision-making.

End of Exam

Advanced Machine Learning 2024
Final Exam
Chennai Mathematical Institute

26 November 2024, 09:30–12:30 (3 hours) **Marks: 50**

Instructions:

- This question paper consists of a total of 9 questions worth 50 marks. All questions must be answered; there are no optional ones.
- Make sure to write you name and roll number on the main answer-sheet *and* on all the additional sheets that you use.
- If you need to use the restroom, ask for a toilet pass from the invigilators.
- Attendance will be taken during the exam. Please sign next to your name on the attendance sheet.
- **NOTE:** Your answers must be in mathematical notation, instead of english, as much as possible. You must properly define any mathematical terms you write. Your answers must be precise, complete and to the point. Otherwise, marks will be deducted.
- **NOTE:** Your handwriting must be clear and legible. Answers written in poor and hard-to-read handwriting will be marked as incorrect.
- **WARNING:** CMI's academic policy regarding cheating applies to this exam. Any form of copying or use of electronic devices is strictly prohibited. Violation will result in an automatic F grade, immediate exclusion from placement activities, and potential expulsion without exception.

formula

1. Let X be the space of all input instances and D is a distribution on X. Let $C \subseteq X$ be a concept class, and let $S \subset X$ be a sample set drawn by sampling from D. Let h be a hypothesis. Then describe what is the training-error and true-error of h? *(5 marks)*

2. Explain why we need gated Recurrent Neural Networks such as LSTM. Why do they perform better than standard RNNs? *(5 marks)*

3. Suppose that you are designing a neural network for some task, such as image classification. You have 3 choices for the depth of the neural network:

   (a) Shallow, i.e. around 5 layers

   (b) (Moderately) Deep, i.e. around 50 layers

   (c) Very Deep, i.e. around 500 layers

   Which of these networks would be the best choice, considering both training and inference? Please justify your answer *(5 marks)*

4. Recall that we studied 2D convolutions in class. Can you describe a 1D convolution of kernel size 5 with 4-channel input. Write down the complete mathematical equation for this convolution. *(5 marks)*

5. Describe the training of Recurrent Neural Networks using Back-Propagation Through Time. *(5 marks)*

6. Describe the Bellman Optimality Equations for an Markov Decision Process *(5 marks)*

7. Explain the effects of the learning rate during the training process when it is *(5 marks)*

   (a) too high

   (b) too low

random

8. Describe the loss-function of the Discriminator and the Generator of a GAN. *(5 marks)*

9. Suppose that you have to build a neural network that takes black and white images as input, and produces color images as output. This can be used to turn old-black and white images to color images. Describe:

   (a) How will you construct the training set for this task? *(3 marks)*

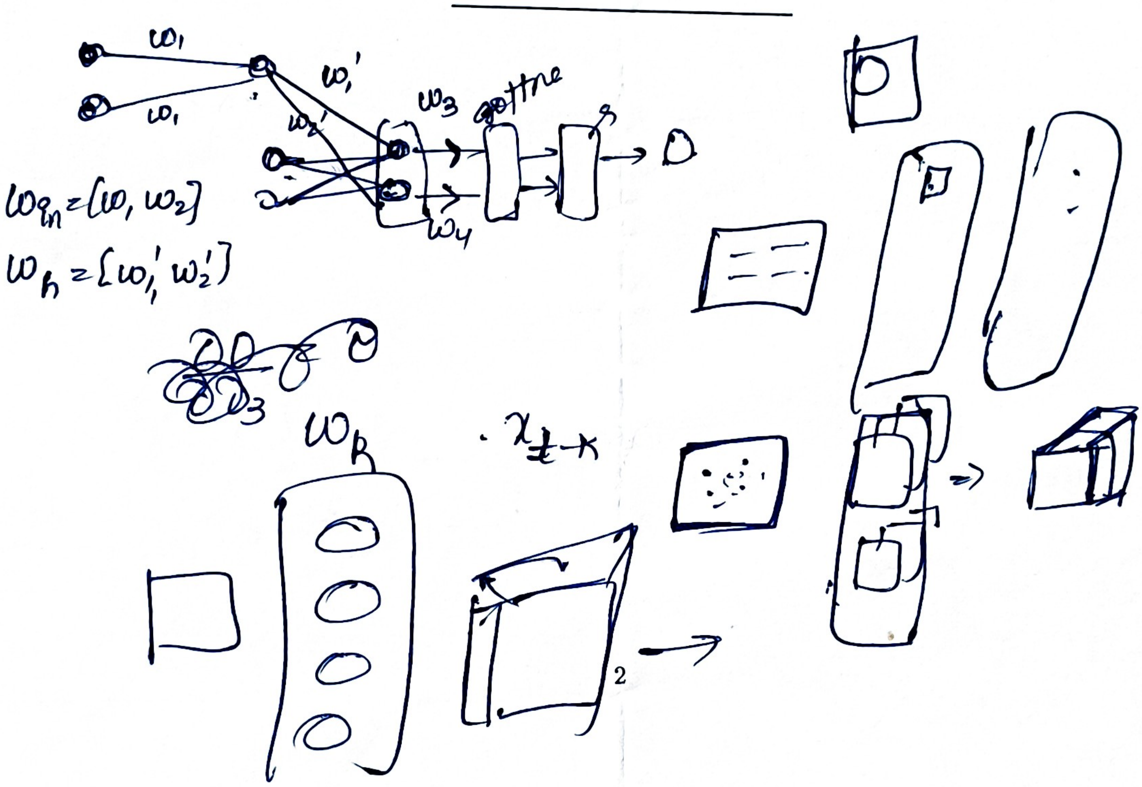
   (b) The architecture of the neural network you will use for this task. *(3 marks)*

   (c) The loss function *(2 marks)*

   (d) You observe that, when your neural network is applied to old black-and-white photos not in the training set, sometimes the output image has errors which could of many different types. Describe one concrete reason for this. *(2 marks)*

Please give specific and detailed answers.

~~[illegible]~~ Aniket Tiwari



16. (1 mark) In the word embedding (computation of word vectors) algorithm, what information is used to determine semantic axis of any word in the semantic space?

1. How frequently a word appears with other words determines the semantic relationship
2. Words that frequently co-occur are placed closer together in the high-dimensional space
3. Co-occurrence statistics is used to compute the word similarity

0 A. Only 1 B. Only 2 C. Only (1) and (3) F. Only (2) and (3) options
3 D. Only (1) and (2) E. Only G. (1), (2) and (3) H. None of the

17. (1 mark) Let us assume that we had used two different unweighted context windows (cw_1 and cw_2) of $size < 7$, (say 3 and 7), to generate two different sets of word vectors for the same corpus. What will be the result of cosine similarity between $\overrightarrow{w_i^{cw_1}}$ and $\overrightarrow{w_i^{cw_2}}$?

1

A. The angle between $\overrightarrow{w_i^{cw_1}}$ and $\overrightarrow{w_i^{cw_2}}$ would be large

B. The angle between $\overrightarrow{w_i^{cw_1}}$ and $\overrightarrow{w_i^{cw_2}}$ would be very small

C. The angle between $\overrightarrow{w_i^{cw_1}}$ and $\overrightarrow{w_i^{cw_2}}$ would be 0 deg

D. None of the above

0 18. (1 mark) HAL algorithm is sensitive to the directions of capturing the context
A. True B. False

1 19. (1 mark) The number of OHVs is equal to the number of tokens in the corpus A. True B. False

1 20. (1 mark) Semantic similarity beween two words (w_x, w_y) is a function of how frequently they appeared in similar linguistic contexts A. True B. False

6. (1 mark) Assume that W_1 and W_2 appeared together a few times (say 2 times) in a corpus. After computing the word vectors, $\overrightarrow{W_1}$ and $\overrightarrow{W_2}$ are found to be close to each other in the semantic space. What could be the possible reason? Mark the closest answer

A. Insufficient Information to make any decision

1 B. Possible only when both words appeared in similar contexts

C. Possible irrespective of the frequency of their co-occurrence in a corpus

D. The statement that the words are found to be close to each other in the semantic space is not correct

0 7. (1 mark) The words appearing in similar contexts (Pick the closest answer)
A. have no relationship B. are similar C. are antonyms D. None of the above

1/2 8. (1 mark) The ideal word vector of a word should reasonably represent A. All synonyms B. Is-A relationship C. Has-relationship D. Consistently occurring neighborhood words

1 9. (1 mark) Select all options that represent the properties of a one-hot vector (OHV)
A. Every word in the vocabulary is represented in an orthogonal axis B. The dot product of any two vectors is zero C. The number of elements in the OHV is equal to the size of the vocabulary D. OHVs capture semantic relationship

1 10. (1 mark) The position of the 1 in the one-hot vector represents the index of the word in the vocabulary list
A. True B. False

6 5/4

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